Cheatsheet AZ-305 — SQL Server on VMs (IaaS): Always On, DR and migration

Applies to **SQL Server installed on virtual machines** (Azure or on-premises). You manage the engine and build HA/DR with the product's technologies. **"Always On"** is the brand of **two** different technologies: **FCI** and **Availability Groups**. They are not the same thing.

1. The technologies in one table

	FCI	Synchronous AG	Asynchronous AG	Distributed AG	Log Shipping	Azure Site Recovery
Protects	Whole instance	Database(s)	Database(s)	A whole AG	Database	Whole VM
Data copies	**1** (shared)	N	N	N	2+	2
Storage	**Shared** (S2D, shared disks, Premium Files)	Own per replica	Own per replica	Own per AG	Own	Own
Replication	—	Synchronous	Asynchronous	Asynchronous	Log backups in batches	Asynchronous (blocks)
RPO	0	**0**	Seconds	Seconds	Minutes	Minutes
RTO	Sec-min	Seconds	Minutes	Seconds	Minutes	Minutes
Failover	Automatic	Automatic	Manual	Manual	Manual	Orchestrated (Recovery Plan)
Readable secondaries	No	Yes	Yes	Yes	Standby only	No
Cross-version	No	No	No	**Yes** (to a higher version)	**Yes**	—
Cross-platform (Win↔Linux)	No	No	No	**Yes**	**Yes**	—
Cluster manager	WSFC	WSFC / Pacemaker	WSFC / Pacemaker	One per AG	No	No
Reach	Local	Local / zones	Regions	Regions / migration	Regions / migration	Regions
Main use	Instance HA	HA with RPO 0	DR	DR and **migration**	Cheap DR and migration	VM DR without touching SQL

2. Always On Failover Cluster Instance (FCI)

- Two or more WSFC nodes; **a single active instance** on **a single shared storage**.
- Protects **everything**: databases, logins, SQL Agent jobs, configuration.
- No replicas → if the storage gets corrupted, there is no copy. Combined with AG for DR.
- In Azure: Storage Spaces Direct, Azure Shared Disks or Premium File Shares as shared storage.
- Signals: "instance-level", "shared storage", "logins and jobs must fail over", "no replicas".

3. Always On Availability Group (AG)

- Each replica has **its own copy** of the data; the primary sends the log to the secondaries.
- Protects **databases** (one or several, as a group). Logins and jobs are **not** replicated (Contained AG in 2022 improves this).
- **Synchronous:** RPO 0, automatic failover, same region/zones. Max. synchronous replicas limited.
- **Asynchronous:** RPO seconds, manual failover, works for another region.
- Up to **8 secondaries**, readable and usable for backups.
- **Listener:** virtual IP/name → in Azure = **Internal Load Balancer + Floating IP + TCP probe** (not HTTP).
- CLUSTER_TYPE = NONE : AG without cluster manager, read replicas only, no automatic failover.
- Signals: "readable secondaries", "no shared storage", "listener", "RPO zero", "several DBs together".

4. Distributed Availability Group

- **AG of AGs:** links two independent AGs (each with its own cluster). Always asynchronous between AGs.
- Supports a **higher version at the target** (2016 → 2019, 2017 → 2022…) and **Windows → Linux**.
- Method documented by Microsoft for **migrations with minimal downtime**: synchronize, manual failover, cutover in seconds.
- Typical target preparation: single-node AG with CLUSTER_TYPE = NONE .
- Signals: "source already has an AG", "migrate to newer version / to Linux", "minimize downtime".

5. Log Shipping

- Periodic log backup on the primary → copy → restore on the secondary (NORECOVERY or STANDBY).

- Simple, cheap, cross-version and cross-platform. Secondary readable only in STANDBY between restores.
- Cutover: last log backup + restore WITH RECOVERY + redirect the app → **minutes**.
- Signals: "no AG at the source", "simple solution", "minutes of downtime acceptable", "low-cost DR".

6. Azure Site Recovery (ASR)

- Replicates the **whole VM** (OS + SQL + everything) to another region; orchestrated failover with Recovery Plans.
- Knows nothing about SQL: RPO of minutes, no readable replicas. Minimum cost (no active VM at the target).
- Signals: "replicate the VM", "relaxed RPO (hours)", "minimize costs", "without touching SQL".

7. Common combinations

Design	Local HA	DR
FCI + asynchronous AG	FCI (instance)	AG to another region
Synchronous AG + asynchronous AG	Synchronous replicas across zones	Remote asynchronous replica
AG + Distributed AG	Local AG	Second AG in another region
Single VM + ASR	None (or Availability Set)	ASR

8. SQL VM rules that show up on the exam

- **Disk host caching:** Log = **None**, Data = **ReadOnly**. If a single policy is required → **None**.
- **OS disk:** Premium SSD **v1** (v2 is not valid as an OS disk).
- **Series with "s"** = Premium storage (DS, Es, Fs…). SR-IOV = Accelerated Networking.
- **SLA:** single VM 99.9 %, Availability Set 99.95 %, Availability Zones 99.99 %.
- **Listener with ILB:** enable **Floating IP**; **TCP** probe on the listener port.

9. Migration tools (summary)

Need	Tool
Assess compatibility beforehand	Data Migration Assistant (DMA)
Migrate to Single DB tolerating downtime	BACPAC
Migrate to MI / VM with minimal downtime (no AG at the source)	Azure Data Studio + SQL Migration extension / DMS online
Migrate 50 DBs to MI offline	Azure Database Migration Service
Migrate from an AG to a higher version or to Linux, minimal downtime	**Distributed AG**
Simple migration, minutes of downtime	Log shipping
Move the whole VM	ASR / Azure Migrate
Oracle/MySQL/DB2 source → SQL	SSMA
Continuous on-prem ↔ Azure connectivity (not migration)	Self-hosted Integration Runtime (Data Management Gateway)

10. Decision tree

```
SQL Server on a VM: what do I need?
├─ Protect the whole instance, with shared storage ..... FCI
├─ Local RPO 0, readable replicas ..... Synchronous AG
├─ DR to another region from an AG ..... Asynchronous AG
├─ Migrate from an AG to another version/platform, minimal downtime  Distributed AG
├─ Cheap DR or simple migration with no AG at the source ..... Log shipping
└─ Replicate the VM without touching SQL, relaxed RPO, minimum cost  Azure Site Recovery
```

11. Signals → answer

Wording	Answer
"shared storage", "entire instance", "logins and jobs"	FCI
"readable secondaries", "no shared storage", "listener"	Availability Group
"failover without data loss" (local)	Synchronous AG
"existing AG" + "newer version / Linux" + "minimize downtime"	Distributed AG
"no AG in source" + "simple" + "minutes of downtime OK"	Log shipping
"RPO 24 hours" + "minimize costs" + "VM"	Azure Site Recovery
"ILB as listener"	Floating IP + TCP probe

12. Mistakes to avoid

1. Confusing FCI (1 copy, instance) with AG (N copies, databases).
2. Ruling out Distributed AG because it is Windows → Linux: **it is supported**.

3. Choosing multi-region AG when the RPO is hours and minimum cost is required → ASR.
4. HTTP probe on the SQL listener → it must be **TCP**.
5. Choosing Session persistence on the listener's ILB → does not apply; what is needed is **Floating IP**.